Machine Learning and IRB Capital Requirements: Advantages, Risks, and Recommendations[*]

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Abstract

This survey proposes a theoretical and practical reflection on the use of machine learning methods in the context of the Internal Ratings Based (IRB) approach to banks' capital requirements. While machine learning is still rarely used in the regulatory domain (IRB, IFRS 9, stress tests), recent discussions initiated by the European Banking Authority suggest that this may change in the near future. While technically complex, this subject is crucial given growing concerns about the potential financial instability caused by the banks' use of opaque internal models. Conversely, for their proponents, machine learning models offer the prospect of better measurement of credit risk and enhancing financial inclusion. This survey yields several conclusions and recommendations regarding (i) the accuracy of risk parameter estimations, (ii) the level of regulatory capital, (iii) the trade-off between performance and interpretability, (iv) international banking competition, and (v) the governance and operational risks of machine learning models.

Keywords: Banking; Machine Learning; Artificial Intelligence; Internal models; Prudential regulation; Regulatory capital

Classification JEL: G21, G29, C10, C38, C55.

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1 Introduction

Within the Basel prudential framework, Internal Rating Based (IRB) models can be used by banks to determine the amount of regulatory capital they must hold to cover unexpected credit losses and ensure financial solvency. Recently, the European Banking Authority (EBA) published two discussion papers (EBA, 2021, 2023) on the use of machine learning (ML) algorithms in the context of IRB models.[1] For the EBA, the objective is twofold. On the one hand, it aims to initiate a discussion on this sensitive topic with the banks.[2] On the other hand, it seeks to define the regulator's expectations concerning sophisticated ML models and to evaluate their compliance with the latest regulations on bank capital requirements.

This subject, though complex, is of primary importance for banking regulators. Indeed, on the one hand, they want banks to embrace innovation and benefit from the latest risk modeling capabilities. On the other hand, regulators need to ensure that these sophisticated models will not become an additional source of systemic fragility (Benoit et al., 2017). Immediately after the publication by the EBA of its first discussion paper, dozens of headlines appeared in the specialized press on the emergence of ML and Artificial Intelligence (AI) in the calculation of banks' regulatory capital. Some commentators mentioned that the EBA has just "opened the door" to ML in regulatory applications. Some have also highlighted the dangers of ML and emphasized the risk of centering poorly controlled modeling within the regulatory framework (Allen, 2021). Conversely, for their proponents, ML models offer the advantage of better measuring credit risk. Thus, they may allow banks to reduce their regulatory capital while ensuring their financial strength, or to facilitate credit access to those whose risks are overestimated by traditional models. Of course, some such optimistic views may reflect financial interests. Indeed, the prospect of ML development in IRB models would generate new business opportunities for consulting firms and the IT providers developing and maintaining these models.

Before exploring the benefits and risks of using ML in the IRB context, we need to high-

[1] The EBA emerged as the first banking authority to publicly express its stance on the adoption of machine learning in the IRB context. For more general discussions about the opportunities, challenges, and regulation of AI in banking in other jurisdictions, see ACPR (2020), Bank of England (2022), Board of Governors of the Federal Reserve System (2021), and Prenio and Yong (2021) for international evidence.

[2] The EBA received responses and feedback from most banking professional associations in Europe, e.g., Asociación Española de Banca, Associazione Italiana Leasing, European Association of Cooperative Banks, European Group of Savings and Retail Banks, French Banking Federation, and German Banking Industry Committee.

light two paradoxes. The first one is that, contrary to what has often been written, the EBA did not "open the door" to ML in 2021; it was already wide open. Indeed, in the Capital Requirements Regulation (CRR) and Capital Requirements Directive (CRD) that govern the use of internal models in Europe, nothing prevents the use of ML algorithms in estimating the credit risk parameters, i.e., the Probability of Default (PD), Loss Given Default (LGD), and Credit Conversion Factor (CCF). However, in practice, only a handful of banks currently use ML in the IRB context, and when they do, these techniques are often confined to data preprocessing or to challenger models. The lack of popularity of ML in the regulatory area of credit risk modeling is surprising as ML is otherwise widely used in the banking industry (ACPR, 2018, 2020; EBA, 2020; Bank of England, 2022), in contexts such as marketing (understanding customer needs), commercial activities (churning, cross-selling), customer relations (banking assistant, robo-advisor), asset management, market risk management, fraud detection, Anti-Money Laundering (AML) and terrorist financing, etc. ML is also widely used in credit risk management whether in the contexts of loan granting and pricing, credit monitoring, recovery or restructuring (IIF, 2019, 2022).

The second paradox is that the original EBA's discussion paper was published less than a month after the publication of the CCR3/CRD6 banking package (October 2021), which implements the Basel IV reform in Europe. This banking package was precisely aimed at reducing the degree of freedom given to modelers in the context of the IRB approach. In particular, the Basel Committee introduced various mechanisms to limit the reduction and the variability of the capital associated with the use of internal models.[3] The CCR3 reform also aimed to restrict the use of approaches based on internal models, with for example the removal of the Advanced Measurement Approach for operational risk. At first glance, it may seem paradoxical for the EBA to engage in a discussion with banks on the use of ML techniques, which are even more flexible than usual parametric models. Indeed, this greater flexibility in the functional forms of models could exacerbate the variability of risk parameter estimates and, ultimately, of the regulatory capital requirements. Given the apparent lack of internal coherence in the regulatory framework – with two concomitant and contradictory messages – it seems increasingly crucial to undertake a thorough investigation of this issue.

[3]For instance, the banking package of Basel IV introduced (i) capital floors so that the level of capital calculated by internal models is not less than 72.5% of the requirements calculated in the standard approach, and (ii) floor values on the prudential parameters of PD, LGD and CCF that enter into the regulatory formula to determine capital requirements.

The objective of this survey is to propose both an academic and practical reflection on the challenges of using ML methods in the specific context of the IRB approach. First, it is important to define ML algorithms and models and to discuss their possible use as internal risk models. Simply put, an ML algorithm is a procedure that permits learning from data, while an ML model is the result of applying this algorithm to new data. The definition of ML adopted by the European regulator refers to model interpretability: A model is understood as an ML model if it is not or hardly interpretable. The notion of interpretability here refers to the degree to which a human being can understand how the model works and how it makes its decisions. In the context of IRB modeling, an ML model can be used (i) to estimate different risk parameters, such as PD, LGD or CCF, and (ii) at different stages of their estimation: data preparation, variable preprocessing or feature engineering, variable selection, or model validation. However, the main challenges arise chiefly when ML methods are used as the primary model for risk differentiation and calibration.[4]

Our conclusions and recommendations can be summarized in three points. The first concerns the *data* used by credit models. Indeed, the question about the use of ML in the context of the IRB approach cannot be dissociated from the data on which these algorithms are trained. In its discussion papers, the EBA explicitly mentions the use of big and/or unstructured data, to which we refer here as "new data". The novelty here is understood in contrast to the usual data employed in bank scoring models, which mainly consist of socioeconomic data and the borrower's payment history, or to the characteristics of the loan contract. Outside the regulatory perimeter, it has been shown that new sources of information can improve risk differentiation, whether for individual borrowers or companies, particularly in the context of the loan granting process. The combination of ML algorithms and new data enables the selection of new default predictors and the identification of risks that may have been ignored by traditional approaches. It may also improve financial inclusion by allowing, for example, better risk assessment of sub-populations of borrowers with little or no credit history, such as young borrowers or newly created companies.

However, the expansion of ML applications to IRB models poses several challenges in terms of data. The collection and management of the data needed to train these ML models must

[4] *Risk differentiation* refers to the process of categorizing borrowers or credit exposures into different risk segments or classes based on their creditworthiness or default probability. *Risk calibration* involves assigning appropriate risk ratings or scores to different risk segments or classes determined during the risk-differentiation process. *Risk validation* is the process of assessing and verifying the accuracy and effectiveness of credit risk models, methodologies, and processes.

comply with the constraints imposed by the CRR. For example, how can we ensure the availability of a 5-year history to comply with the CRR when these new data usually do not have such historical depth? Moreover, as internal models constitute the core of banking activities, banks are reluctant to mobilize data or scores produced by external providers to estimate PD, LGD or CCF. All these constraints raise the question of whether it is worth discussing the value of this new data in the IRB context. However, the aim of ML expansion is to permit the banks to remain as close as possible to the technological frontier and to prevent a technological oligopoly from forming to the benefit of only a few players. Indeed, the question of the use of ML in the IRB context touches on issues of international banking competition and barriers to entry for new players such as fintechs or bigtechs. Regulating ML use within the IRB framework is thus crucial if we want to avoid an actor being able to take advantage of privileged access to data to distort banking competition.

Our second conclusion concerns the potential benefits of ML in terms of *statistical accuracy* and *regulatory capital*. Among practitioners, it is well accepted that ML models provide more accurate estimates of risk parameters than usual parametric approaches, such as logistic regression. However, one should maintain a sense of caution in this matter. For the modeling of PDs, the academic literature clearly shows that (i) only ensemble methods (particularly Bagging and Boosting) significantly improve risk parameter estimates compared to traditional parametric approaches and that (ii) these predictive gains tend to plateau quickly with increasing model complexity. For example, recent studies have shown that in the context of credit scoring, deep learning does not provide a significant gain compared to the ensemble methods used to date (Alonso and Carbó, 2020; Gunnarsson et al., 2021). These results can be explained by the fact that, unlike other ML applications (e.g., biomedical, environmental), the assessment of a borrower's creditworthiness is a relatively simple problem in which nonlinearities and interactions between explanatory factors are not very strong. Hence, the accuracy gains of ML mainly come from a reduction in the variance of estimates through the aggregation of weak predictors (Bagging methods such as Random Forests) or an iterative approach of over-weighting of prediction errors (Boosting methods such as XGBoost). This explains why these two methods are so popular among banks.

Improving the accuracy of banks' estimates of Basel risk parameters does not always lead to lower regulatory capital (Alonso and Carbó, 2020; Fraisse and Laporte, 2022). To illustrate this point, let us suppose that replacing logistic regression with an XGBoost model as the primary model for PD differentiation leads to better performance, such as a higher area under the receiver operating characteristic curve (AUC). Nothing guarantees that this improvement will decrease the estimated PDs for all or some of the loans in the portfolio. Indeed, the model's improved predictive ability does not necessitate that the logistic regression led to an overestimation of the PDs. Moreover, the estimated PDs can decrease for loans with low Exposure At Default (EAD) or low LGDs, yet increase for others. Therefore, an ML method can perfectly reduce the AUC of a PD model or the R^2 of an LGD model, without necessarily leading to a decrease in the level of Risk-Weighted Assets (RWA).

Our third conclusion addresses the issue of *interpretability*. While ML models can improve the accuracy of risk parameter estimates, this improvement in precision often comes at the cost of a more complex relationship between the input and output variables of the model. The trade-off between estimation accuracy and complexity/interpretability is key to deciding whether to use ML models for prudential purposes. The challenge here is to articulate the relationship between ML models and human judgment concerning risk differentiation, which clearly raises questions about the regulator's interpretability requirements. The answers to such questions involve two opposite visions. The first consists of using non-interpretable ML models and post hoc techniques that allow us to interpret and explain their forecasts (e.g., PDP, LIME, SHAP). This two-step approach of estimation then interpretability is usually chosen by banks and is discussed at length in the EBA discussion papers. However, this approach also raises concerns (Rudin, 2019). The main difficulty is that there is no universal method of interpretability and each of these methods refers to a particular explanation of how the model works. Therefore, an explanation taken individually may not fully meet the expectations of a modeler, internal auditor, or supervisor. Furthermore, these different methods can provide divergent (Krishna et al., 2022) and/or unstable explanations (see Chen et al. (2022) for an application on loan portfolios with few defaults).

The alternative vision, which we advocate in this paper, is to use ML models that combine high performance and native interpretability. This approach challenges the notion, widely accepted in the industry, of an inherent trade-off between predictive performance and interpretability. Recent theoretical work based on the concept of Rashomon space has shown that for any classification or regression problem, there is very likely to be a natively interpretable model that performs as well as the best black box model (Semenova et al., 2022). However, even if this theoretical result is proven, some may think that it is always easier in practice to implement a black box model than to search for a hypothetical model that both performs well and is interpretable. On the contrary, several examples have illustrated that it is possible to build high-performance, natively interpretable models (Dumitrescu et al., 2022; Flachaire et al., 2022), and natively interpretable ML is currently used and promoted by several American banks (Sudjianto and Zhang, 2021). Such an approach could clearly constitute a development path for ML in the IRB context.

In addition to interpretability, the use of ML algorithms raises the question of specific governance allowing for the assessment of all risks, from the construction phase to the implementation phase. Finally, another challenge for banks and their supervisors lies in the operational risks associated with the implementation of ML models, such as those related to the setting of hyperparameters and overfitting.

2 Is ML currently used in the IRB context?

In this section, we define the concept of an ML model and discuss its uses within the IRB approach. We show that these models are currently rarely used by banks for regulatory applications (e.g., IRB, IFRS 9, regulatory stress-tests), even though they are widely used in other non-regulatory contexts of credit risk analysis (e.g., loan granting, risk monitoring).

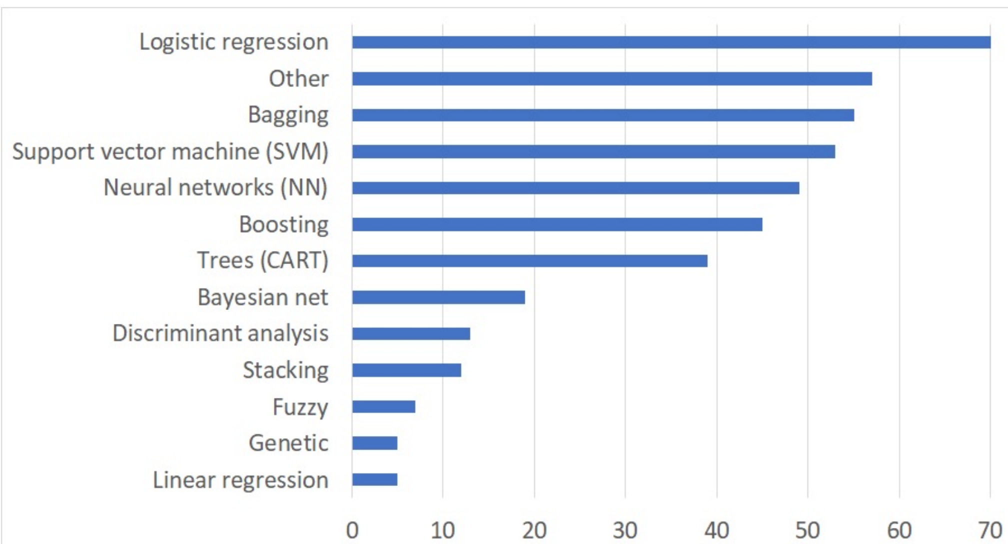
2.1 Defining ML in the IRB context

In a recent report on the use of big data in banking, EBA defined ML as "*a subcategory of AI that uses algorithms able to recognize patterns in large amounts of data via a learning process in order to make predictions based on similar data. The learning is done by means of suitable algorithms, which are used to create predictive models, representing what the algorithm has learnt from the data in order to solve the particular problem*"(EBA, 2020). According to this general definition, most statistical methods and econometric models used by banks fall under the umbrella term of ML. For example, the logistic regression can be assimilated to ML, as can classification or regression trees, which a priori do not pose any problem in terms of interpretability.

However, in the IRB context, the EBA limits the definition of ML "*to models that are characterised by a high number of parameters and, therefore, require a large volume of (potentially unstructured) data for their estimation that are able to reflect non-linear relations between the variables.*" (EBA, 2021). This definition includes all noninterpretable black box type models, as well as all types of complex models.

The notion of *model complexity* in ML remains ambiguous and there is an abundant literature devoted to its quantification. Complexity can be assessed according to different theoretical approaches such as the Vapnik-Chervonenkis theory (Vapnik, 2013) or the Ockham's razor. If we limit ourselves to the simple case of a decision tree, complexity can be measured by its depth and the number of terminal nodes. For instance, a shallow decision tree would not fall under the definition of ML in the IRB context as per the EBA. Differently, a very deep tree with many terminal leaves would be classified as ML as it requires the estimation of numerous parameters and is difficult to understand due to the complexity of the induced decision rules. Thus, it is the algorithmic complexity and lack of interpretability that define the notion of ML in the specific context of IRB models.

Figure 1: The most popular ML methods for credit scoring



Note: This figure displays the number of times a given ML algorithm has been used in the 110 articles considered in the survey conducted by Markov et al. (2022). Each article can use more than one algorithm. Source: Markov et al. (2022).

It is virtually impossible to compile an exhaustive list of all the ML algorithms actually used for credit risk modeling, even in non-regulatory applications. For example, Lessmann et al. (2015) list no fewer than 41 different classification algorithms used for PD modeling, in 48 academic articles published between 2003 and 2015. In a more recent study, Markov et al. (2022) analyze 110 articles published between 2016 and 2020 to identify the ML algorithms, evaluation criteria, and most commonly used preprocessing methods in the academic literature on credit scoring. As shown in Figure 1, apart from logistic regression, the most commonly used ML methods are Bagging methods, Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Boosting methods.[5] For LGD modeling, the most commonly used algorithms appear to be regression trees, Random Forests, Gradient Boosting methods, ANN, and SVR (Loterman et al., 2012; Yao et al., 2017).

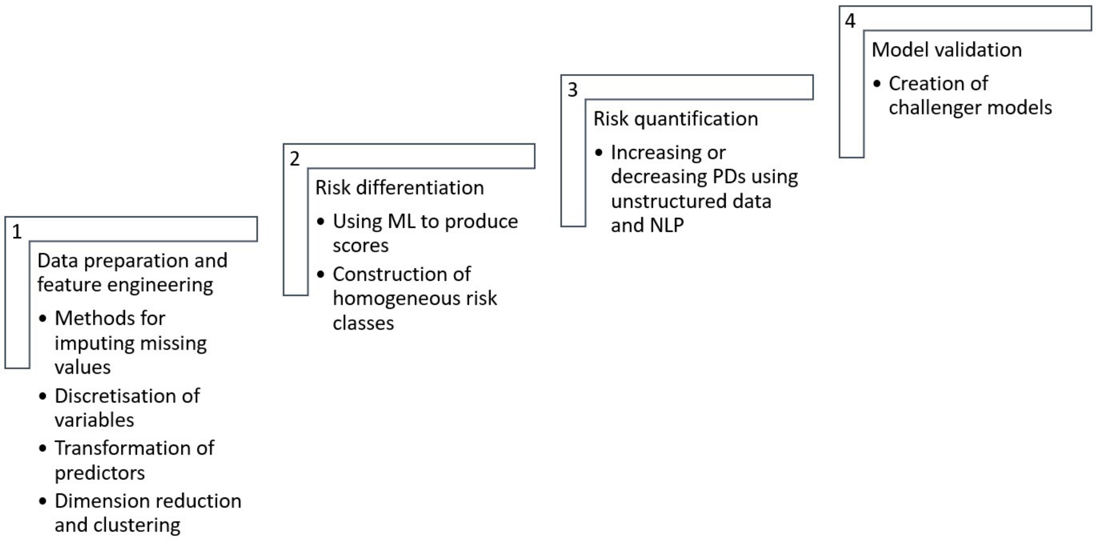
2.2 Potential ML applications in the IRB context

In the IRB context, ML algorithms can be utilized for the modeling of PD, LGD, or CCF. In PD models for instance, ML can be used at four stages of the modeling process. The first step covers *data preprocessing, feature engineering*, and *feature selection*. The second step, referred to as *risk differentiation*, involves using a model to estimate a continuous score based on risk factors and then categorizing borrowers or credit exposures into different risk segments or classes based on this score. The goal is to group borrowers into homogeneous risk classes, ensuring a uniformity of default risk within a given class and the heterogeneity of this risk across classes. The third step, known as the *risk quantification* step, involves assigning a default probability to each homogeneous risk class. Generally, these probabilities are estimated by the empirical frequency of defaults observed over a long period within each class. These default probabilities are the key inputs for the calculation of capital requirements. The last step, referred to as *validation*, aims to ensure the consistency and stability of the default forecasts from the internal model. The EBA distinguishes different possible applications for ML models in these different phases of IRB modeling, as summarized in Figure 2.

Data preparation and feature engineering. ML algorithms can be used upstream of modeling, in the preprocessing phases of data preparation and feature engineering. These preprocessing phases aim to increase model robustness (e.g., handling missing and outlier values), facilitate the interpretation of model results (discretizing continuous variables), eliminate redundant information (removing some highly correlated variables), preselect the candidate variables for modeling (analyzing correlation with the target variable), and increase model predictive performance (optimizing the binning methods to discretize the continuous features).

[5] Another meta-analysis of credit scoring methods based on 74 articles published between 2010 and 2018, was conducted by Dastile et al. (2020). This study provides slightly different results from those of Markov et al. (2022), since the most used algorithms in their sample are, in decreasing order, the logistic regression, SVM, ANN, decision trees, Boosting, Random Forests, and then other Bagging methods (AdaBoost, etc.).

Figure 2: Examples of ML uses in IRB models



It is well known that feature engineering is instrumental in increasing the predictive capabilities of scoring models (Verdonck et al., 2021). According to Dastile et al. (2020), the most commonly used methods are standard statistical techniques for transforming univariate variables (e.g., Box-Cox transformation for continuous variables, dummy coding, percentile coding, transforming polytomous categorical variables into binary variables) or multivariate variables (e.g., principal component analysis, linear discriminant analysis). However, an increasing number of ML techniques are also used for data preprocessing. A good example is the autoencoder, which is a type of three-layer neural network that can efficiently reduce the dimension of the data representation set. Also within this category are ML techniques used for imputing missing values, such as Bayesian Network Iterative Imputation (BNII) methods, or Bagging methods such as XGBoost and CatBoost (Markov et al., 2022).

Feature selection. One of the main advantages of ML algorithms lies in their ability to select and combine relevant explanatory variables. This capacity constitutes one of the main differences from traditional econometric approaches, in which model specification is based on prior economic reasoning and statistical tests (Mullainathan and Spiess, 2017; Charpentier et al., 2018). In contrast, ML algorithms allow for the automatic construction of a predictive model from the data, without any a priori inference procedure, by selecting and combining explanatory variables. In this regard, ML algorithms are similar to automatic methods of selecting explanatory variables (e.g., stepwise, backward, autometrics), which are commonly used in econometrics. An important difference, however, is that ML algorithms can handle a large number of variables and automatically create new variables by transforming the initial ones. In general, the use of ML in the variable selection phase can be performed concurrently with modeling (e.g., when an ML algorithm is used for risk differentiation) or in a variable preselection approach aimed at reducing the dimension of the classification or regression problem. In the latter case, selection is often performed through penalized regression methods, such as Lasso (Tibshirani, 1996) or Adaptive Lasso (Zou, 2006).

Risk differentiation. The EBA states that the level of attention devoted to ML methods should depend on the types of its applications in the IRB context. Obviously, a lower level of attention is required when ML models are used upstream or downstream of risk differentiation. The most significant issues arise when ML is employed as a primary model to construct a continuous score and allocate credits into homogeneous risk classes. In the rest of the analysis, we will focus on the crucial phase of risk differentiation.

Risk quantification. ML methods are not generally used to calibrate LGD or PD parameters within risk classes, as this calibration is based on long-term observations of default frequencies or losses. Thus, risk quantification does not require any predictive model. However, the EBA mentions potential applications of ML that could be used to carry out adjustments of PDs based on textual analysis (e.g., financial reports).

Model validation. One of the most common uses of ML in the IRB context is in the model validation phase of risk differentiation. In practical terms, this often involves developing "challenger models", which are used as benchmarks to evaluate the risk parameter estimates (or the RWA and the regulatory capital requirements) provided by the internal models in production. Since these challenger models are not used directly for calculating regulatory capital requirements, they escape the constraints imposed by the CRR. Hence, this is the easiest application of ML in the IRB framework, particularly in terms of governance and regulatory compliance.

2.3 Actual ML applications in the IRB context

Do banks actually use ML in the IRB context? Some aspects of the answer to this question can be found in two surveys recently conducted by the Institute of International Finance (IIF) on the use of ML in the credit risk industry (IIF, 2019, 2022).[6] As shown below,

[6]The first survey (IIF, 2019) questions representatives of 60 international banks on their use of ML for credit risk analysis between 2018 and 2019. The sample includes 10 banks from the US, 4 from Canada, 14 from the eurozone, 7 from the rest of Europe, and 9 from Asia. The second survey (IIF, 2022), conducted

both surveys confirm the paradox of ML in credit risk management: it is widely used by international banks in managing their credit risk outside the regulatory perimeter, but only rarely in the regulatory context.

ML for Unregulated Credit Risk Management. The IIF (2022) survey shows that 52% of sample banks use ML models in production for credit risk management, while another 25% of banks are currently experimenting with these techniques. Only 9% of the banks surveyed do not use ML and do not plan to do so. However, these aggregate figures mask some important geographic differences. ML is widely used by banks in the eurozone and the United States, where the adoption rate in production is 100%.

The IIF surveys also show that most banks that use ML have been doing so for a long time and in several functions of the credit risk process, i.e., data cleaning, variable selection, data exploration and segmentation, model development, and model validation. Finally, ML is used for all types of portfolios although the highest adoption rate is found primarily in the context of retail lending.

ML for Regulated Credit Risk Management. While ML methods have been used efficiently by banks for more than a decade, they are still very rarely used in IRB models. Only 10% of the banks surveyed state that they use ML in production at some stage of their IRB modeling (IIF, 2019). Even worse, the use of ML has been declining since 2018. If we include banks stating that they plan to use ML for the calculation of RWA, the rates are 15% in 2019 compared to 23% in 2018. These ML usage rates can be compared with those declared by banks in 2019 for their credit granting models (75%), risk monitoring models (60%), and restructuring and recovery models (26%). Furthermore, for these three types of non-regulatory models, the trend is positive between 2018 and 2019. Interestingly, the low penetration rate of ML in IRB models also affects other regulatory topics, such as stress tests (10%) or provisioning (8%).

It is obvious that banks have sufficient knowledge and experience to develop ML models in the IRB approach. They are already widely using this technology in other areas that do not require any regulatory approval (ACPR, 2018). Therefore, as indicated by the responses from banking professional associations to the EBA consultation, it is primarily a

in 2022 on a sample of 43 international banks, focuses on the uses of ML for both credit risk and AML. The sample includes 8 banks from the eurozone, 7 from other European countries, 3 from Latin America, 4 from the US, 2 from Canada, 4 from Japan, 3 from China, and 5 from the Middle East or Africa.

problem of regulatory views on the acceptability of these models that limits their application. Knowing that the development of IRB models is a process that requires considerable time and resources, banks appear to be waiting for a clearer approval process for ML models before investing in and deploying them at large.

3 Which data for ML-based IRB models?

Because of their very definition, ML models cannot be dissociated from the data on which they are trained (learning sample) or evaluated (test sample). Thus, when discussing the contribution of ML in the context of internal models, we must distinguish what comes from ML techniques and what comes from the data. Among all the data available, a key role is now played by the big, new, or unstructured data, of which ML permits exploitation. In this section, we discuss this new type of data, its potential contribution, and the issues it raises in terms of governance and banking competition.

3.1 "New data"

The idea that ML allows risk modelers to exploit new data to improve risk differentiation is not specific to the IRB context. Rather, this idea lies at the center of the broad methodological debate on credit scoring. Today, there is consensus that ML credit scoring models have plateaued in terms of forecasting performance. Consequently, the only way to improve these models is now by exploiting new sources of data (Oskarsdottir et al., 2019), which raises the question of what exactly these new data sources are. The novelty here must be understood in contrast to the traditional data used in credit scoring: information concerning the borrower's characteristics (age, income, marital status, etc.), banking history, or the features of the loan. These new data, or "data-diversity", can originate from open banking (revised Payment Services Directive or PSD2), the digitization of customer relations, social networks, etc.

Of importance to policy is the extent to which these new data can capture weak signals of default and, ultimately, promote access to credit for individuals or businesses often considered too risky by traditional models. In the context of credit scoring models, several studies show that these new data allow us to reveal weak signals that significantly improve the quality of the assessment of loan applicants' creditworthiness. A first useful comparison is provided by Jagtiani and Lemieux (2019) who show that the correlation between the proprietary scores of the LendingClub credit platform and the FICO scores obtained for a comparable loan sample diminished from 0.80 for loans contracted in 2007 to less than 0.35 for loans contracted in 2015. Similarly, Berg et al. (2020) analyze the credit score of an e-commerce company that uses digital footprint data left by online customers, such as the type of device used, operating system, type of email address, and time of connection. They show that this information improves the performance of the scoring model compared to a model based only on the customers' socioeconomic data. More recently, Rishabh (2023) also documents the over-performance of models based on payment footprints compared to credit scores and suggests that payment footprints could even substitute credit scores. Using a rich set of loan application data from the largest German fintech lender in consumer credit, Nam (2022) studies the effects of open banking and customer-driven data sharing. She shows that high-risk borrowers are more likely to share their bank account data than low-risk borrowers. Agreeing to share open-banking data permits applicants to increase their probability of loan approval by 9 percentage points and lowers interest rates by 2 percentage points.

Another example is given in the data-heavy study by Oskarsdottir et al. (2019). They demonstrate that mobile phone data are a rich source of information for assessing the default probabilities on consumer loans, particularly for the most vulnerable borrowers. The data in their study, jointly provided by a bank and a telecommunications company, include the call logs of the bank's customers, the banking history, and the sociodemographic data of these customers. The study covers one and a half years of banking history for over two million bank customers and records the call activity of nearly 90 million unique telephone numbers over a five-month period. These call logs are used to build call networks, which are used to compute various statistical indicators (e.g., link indicators, PageRank-like measures). The latter are then used as explanatory variables to improve the predictive performance of a default model applied to credit-card users. The findings clearly indicate that classification models built with features representing call behavior perform better, both in terms of AUC and economic performance, than models based solely on banks' customer data and socio-demographic data. Oskarsdottir et al. (2019) conclude that this type of information could be used in a positive way to improve access to financing for individuals whose credit would potentially have been refused based on traditional information (e.g., young borrowers without a banking history).

These new data can be collected and processed directly by banks, but they can also be obtained from fintechs in the business of building and selling scores to banks (vendor models or scorecards). These scores can be used directly or as additional input in their own scoring models. An example here is the fintech Big Data Scoring, a service that enables banks to integrate social media data related to the company and/or its manager into their lending models. These scores are sometimes built to offer banks new tools to develop financial inclusion. For instance, the fintech Zest AI builds scores from highly disparate information to assess the solvency of young clients with very limited or no credit history. The use of these new data sources and ML then allows for increasing acceptance rates, while controlling risks and potentially limiting discrimination biases.

Given the regulatory constraints in place, most of the empirical investigations conducted in the credit scoring domain cannot be implemented in the IRB domain. Specifically, the CRR requires that the data used in IRB models be accurate, complete, and appropriate. Such constraints are often incompatible with the collection, transformation, and use of new data sources. For instance, the CRR mandates that credit institutions estimate PDs by risk class based on long-term averages of one-year default rates and use a historical observation period of at least five years. Similarly, LGD estimates must be based on data covering a minimum five-year period. For many alternative data, such long time series simply do not exist. Furthermore, when the collection and processing of these data are outsourced to a data provider, questions arise concerning how to guarantee the quality of processing and the long-term reliability of external data sources. When using an alternative score produced by a third party, the bank would need to ensure compliance of the new data sources and the processing carried out by the provider, which would further complicate the assessment of the compliance of its IRB models. Such difficulties are probably why, in the responses to the EBA, no European banking association has reported any outsourcing, or even any outsourcing project, in their IRB models by any of their members.

Thus, the only sources of new data that seem exploitable in the IRB approach are either data collected under new environmental regulations, or unstructured data, most often textual data, currently stored by banks, but not yet integrated into internal models (Guégan and Hassani, 2018). One example concerns data related to greenhouse gas emissions, such as Scope measurements, for corporate loans, or energy performance diagnostics (EPDs) for real estate loans. In the case of corporate loans, one could also exploit non-financial reports through Natural Language Processing (NLP), especially since the European Commission recently announced its intention to review the Non-Financial Reporting Directive (NFRD) as part of its strategy to strengthen the foundations for sustainable investment.

3.2 Data and banking competition

If very few banks use, or even plan to use, such new data sources for their internal models, why are banking authorities interested in this topic? An explanation is offered by the implications, in terms of competition, of the use of these data by *new actors* who could disrupt the credit market (Vives, 2019; Boot et al., 2021). Could a new player benefit from an informational advantage and drastically distort banking competition? Such manipulation could be accomplished either through a more efficient augmented credit granting process or through internal models that would reduce credit capital charges.

The question of informational advantage on credit markets is particularly relevant for bigtechs, such as GAFAM (Google, Amazon, Facebook/Meta, Apple, and Microsoft) and BAXT (Baidu, Alibaba, Xiaomi, and Tencent). The cooperation initiated by bigtechs with numerous financial players, their ability to mobilize a very large user base, and their advanced AI expertise suggest a growing influence of these platforms on the financial sector. A first action for bigtechs is to provide an in-house credit offering. A good example is Alibaba, a pioneer in the credit market with the Alipay payment application, which has offered consumer microcredits since 2015. Also notable is the *Apple Card* credit card jointly launched by Apple and Goldman Sachs in 2019. In addition to extending credit to customers, bigtechs have also extended credit to the merchants operating on their platforms (Li and Pegoraro, 2022). Apart from standard features related to creditworthiness, bigtechs can also exploit real-time information about their borrowers' transactions. Moreover, the platform can monetize the information it collects on consumers or merchants, by reselling it to financial service providers. Such new data could be used to improve their credit granting process, risk monitoring, or even their regulatory internal models. Regardless of the credit distribution channel, these platforms can be analyzed as "two-sided" markets (Rochet and Tirole, 2003), connecting consumers and financial service providers. The platform, in a quasi-monopoly situation, is paid by charging entry fees to both sides of the market and receiving commissions on all transactions made (Clerc et al., 2020).

4 What are the expected benefits of ML for IRB modeling?

The implicit idea behind the use of ML in the IRB context is that it makes credit less "expensive" in terms of regulatory capital. This idea is based on two assumptions. First, internal models of PD or LGD based on ML techniques outperform traditional approaches. For instance, ML may better predict default on certain subpopulations of the credit portfolio, by capturing weak signals that standard models cannot capture. Second, these gains in accuracy translate into gains in terms of regulatory capital, while still meeting the regulator's model validation requirements. As evidenced below, although the first assertion can be validated under certain conditions, the second is not necessarily true in practice. In the remainder of the analysis, we will focus on the use of ML during the risk differentiation phase, with a constant information set (no new data).

4.1 Potential gains in terms of accuracy and productivity

Regarding the expected benefits of ML in credit risk management, the top three answers provided by banks in the IIF (2019) survey are (i) the improvement in the accuracy of risk parameter estimates (70%), (ii) the identification of new risk segments (58%), and (iii) some productivity gains in model deployment and data preprocessing (50%).

Improved accuracy of risk parameter estimates. The advantages of supervised ML over traditional parametric methods are well known (Mullainathan and Spiess, 2017). These algorithms can flexibly select the functional form that links the target variable (typically a binary variable associated with default) and the features available in the training dataset. In this respect, ML is similar to semi-parametric econometrics. The main difference is that ML algorithms can be applied in the presence of many predictors. Likewise, ML algorithms autonomously select the variables that enter the model specification and thus resemble automatic methods of variable selection in econometrics. Most importantly, these algorithms can create new predictors by combining and/or transforming the initial predictors. For example, an SVM algorithm allows the projection of a linear separator into a higher dimensional space through a kernel function. This kernel transformation implicitly creates new explanatory variables by transforming the initial variables (Dichtl et al., 2023).

The flexibility of ML models is particularly effective in detecting non-linearities and interactions between predictors. However, to answer whether accounting for these non-linearities and interactions truly boosts model performance, we need to distinguish between two generations of ML algorithms used in credit scoring. The first generation corresponds to so-called "individual" supervised classification methods that aim to partition the feature space to predict the event. The most popular methods in the field of credit risk are classification trees, SVMs, and ANNs. The second generation corresponds to ensemble methods that use an aggregation or combination of many individual learning methods (weak learners) to reduce the variance of the model and avoid overfitting.

Interestingly, credit risk modeling was one of the very first use cases of ML methods. For instance, the CART algorithm (Classification and Regression Tree) was used for a credit scoring application less than a year after the publication of the article by Breiman et al. (1984). Similarly, the first applications of neural networks in this context date back to the early nineties (Tam and Kiang, 1992; Altman et al., 1994). However, it quickly became apparent that these individual classifiers did not significantly outperform logistic regressions (see for instance the survey of Thomas (2000)). In a comparative study based on 17 algorithms and eight databases, Baesens et al. (2003) confirm that SVMs or ANNs deliver very good predictive performance. However, for most databases, the differences between the AUCs of the best ML method and of logistic regression are less than 2%. The main conclusion is that credit scoring is a field with too few non-linearities in the data for the predictive performance gains of ML to be significant. This is probably why the adoption of ML techniques in credit risk models remained low before the early 2010s.

The change came with the use of the first ensemble methods, particularly the Bagging methods (a contraction of the terms Bootstrap Aggregation) introduced by Breiman (1996) and the Boosting methods introduced by Freund and Schapire (1997). Examples of Bagging methods, such as Random Forests and Rotation Forests, rely on bootstrap sampling and aggregation techniques aimed at improving the predictive quality of weak learners, such as decision trees.[7] Under certain conditions, it can be shown theoretically that the combination of these individual classifiers obtained on bootstrapped samples reduces the variance of the aggregated classifier, and thus improves its predictive performance on a test sample. Boosting methods (e.g., XGBoost, AdaBoost) rely on an adaptive strategy to "boost" the

[7]Bagging involves training a set of classification trees on bootstrap resampled subsets, i.e., those containing individuals randomly drawn from the initial sample. The variance reduction obtained is then a decreasing function of the correlation between the individual trees trained on these resampled subsets. The Random Forest method aims to further reduce this correlation by resampling both the individuals in the training set and the variables candidates for binary division.

predictive performances of a weak predictor.[8] The general idea is to iteratively train a model to reduce the forecast errors obtained at each step. Some Boosting methods, such as the XGBoost method (Extreme Gradient Boosting) developed by Chen and Guestrin (2016), are very popular in classification competitions, such as Kaggle.

Dozens of scientific articles have shown that ensemble methods outperform parametric models in forecasting loan default. An early comprehensive study is the one by Lessmann et al. (2015), which evaluates 41 classification algorithms, including many advanced ML algorithms using both statistical and economic evaluation criteria. They clearly show that ensemble methods predict defaults significantly better than a logistic regression. Random Forests systematically dominate individual classifiers (logistic regression or ML). However, the authors also show that the gains in predictive performance provided by ML tend to plateau with increasing model complexity. For instance, the AUCs obtained with Rotation Forests type algorithms do not differ from those obtained with simple Random Forests.

A similar conclusion is reached by Alonso and Carbó (2020), who contrast the predictive gains of ML with the additional costs incurred in terms of supervision. Their literature review shows that beyond homogeneous Bagging and Boosting ensemble methods, methodological developments such as Deep Learning, reinforcement learning, or heterogeneous ensemble methods, produce mixed results and do not significantly improve default prediction. Gunnarsson et al. (2021) also show that Deep Learning algorithms do not seem to be appropriate for credit scoring, as it is often the case for tabular data. Their comparative study carried out on 10 databases, concludes that Deep Learning methods (deep belief and deep multilayer neural networks) are not the most effective, yet are the most costly in terms of computing time compared to ensemble methods (XGBoost).

Notably, these academic evaluations of ML models do not consider the preprocessing that is generally applied by banks. Within the IRB context, the construction of a PD model generally relies on a series of data and feature preprocessing steps that are entered into the logistic regression, which is ultimately used for risk differentiation. For example, continuous variables are generally discretized to limit the effects of extreme values and to facilitate the interpretation of the score grid. Discretization thresholds are sometimes chosen to maximize the model's discriminatory power. Similarly, the modalities of categorical variables are

[8] An AdaBoost classifier is a meta-model that fits a predictor on the original dataset and then trains additional copies of this predictor on the same dataset, but over-weighting misclassified instances so that subsequent classifiers focus more on these prediction errors.

grouped together to help the model better identify the risk of default in the portfolio. Finally, the pre-selection of model variables is based on a series of tests to identify the variables most related to the target (Kruskall-Walis tests, chi-square tests, measurement of Cramer's V) and to remove some highly correlated variables. The final selection of the features included in the logistic regression model then relies on an automatic selection method (stepwise) and on a detailed analysis of the model's predictive results on a test sample. However, in the academic literature, the authors do not reproduce the IRB methodology and the logistic regressions are usually estimated with non-transformed data. Therefore, there is a risk that academic studies overestimate the gains in terms of the accuracy of ML techniques.

Other ML applications in risk modeling. ML can also be used for modeling other risk parameters, such as LGD, CCF, or EAD. To date, most of the academic literature has focused on LGD, and applications to CCF or EAD remain scarce. Most studies on CCF/EAD have considered parametric specifications of models with fractional responses to Papke and Wooldridge (1996) or of a Zero-Adjusted Gamma (ZAGA) type model (Tong et al., 2016). In studies of LGD, the interest in ML lies in its flexibility and ability to capture the multimodal nature of the distribution of losses in the case of default. However, this contribution is much more limited than in the case of PD since the R^2 of the LGD models are often very low, even for ML models (Hurlin et al., 2018). This is why, in practice, LGDs are rarely calibrated using complex models. Instead, banks generally use simple segmentation techniques (contingency tables) that group loans into homogeneous groups and assign them an average recovery rate observed over a long period.

The comparative study by Loterman et al. (2012) evaluates 24 LGD modeling techniques using six databases of LGD observed on consumer loans. The authors compare the performance of parametric models (linear regression, survival model, fractional response regression, Beta regression, Tobit model, etc.) with one of several ML methods (regression trees, Random Forests, Gradient Boosting methods, ANN, and SVR). While all the considered LGD models exhibit a low predictive performance (with R^2 varying between 4% and 43%), SVM and ANN tend to deliver slightly better performance. Several studies also show that two-step approaches combining ML methods and parametric methods better capture the multimodal dimension of the loss distribution (Bellotti and Crook, 2012; Yao et al., 2017).

Improving productivity. Another argument to justify the use of ML concerns the productivity gains that these techniques can generate (Grennepois et al., 2018). Such gains are likely to materialize during the aforementioned data and feature preprocessing phase. For instance, it is acknowledged that the predictive performance of Random Forests is generally robust to the non-imputation of missing values, the presence of strong correlations among some explanatory variables, the non-grouping of the modalities of discrete variables, and the non-discretization of continuous variables. Similarly, it is observed that methods such as Random Forests, Boosting, or neural networks are more robust to class imbalances than decision trees, linear discriminant analysis, or logistic regression (Chen et al., 2022). Therefore, the use of these methods can avoid the need for some of the preprocessing required in the context of logistic regression models. Although the productivity gain argument is probably more relevant for the modeling of credit scores for loan applications or non-regulatory monitoring scores that are frequently revised, it is not to be excluded in the IRB context. Beyond productivity gains, ML can reduce potential modeling biases due to data preprocessing since, in the end, these algorithms let the raw data speak.

4.2 Reduction in regulatory capital

There is reason to believe that banks will be more likely to adopt ML for their IRB model if it permits them to reduce capital requirements (Repullo and Suarez, 2004; Ferri and Pesic, 2017; Pérez Montes et al., 2018; Colliard, 2019). However, it is important to understand that higher accuracy in the estimation of risk parameters (PD, LGD and CCF) does not necessarily imply lower regulatory capital. There are two main reasons for this situation. First, to reduce regulatory capital, the overperformance provided by ML needs to come from lower PDs or LGDs. However, there is no indication that the predictive gains associated with Random Forests or Gradient Boosting techniques are due to an overestimation of default risk by traditional parametric approaches. Second, even if ML methods permit lower regulatory capital, these models still must be validated by the internal validation team and, most importantly, by banking authorities. At present, there are few academic studies on the economic evaluation of ML models in the IRB context. As shown below, no ML model unconditionally dominates its competitors (either parametric or ML) on the three key dimensions of accuracy, validation, and capital requirements/RWA.

PD modeling. The most comprehensive study on the impact of ML on regulatory capital requirements is that by Fraisse and Laporte (2022). Their approach comprises (i) constructing pseudo-internal PD models based on ML algorithms, (ii) verifying that these models would be validated by a supervisor, and (iii) comparing the RWA derived from these models to those obtained from the usual models. The latter typically combine logistic regression and expert judgment. For the empirical analysis, the authors use granular data on six corporate credit portfolios from large French banking groups, which represent 80% of the credits extended to businesses over the period.[9] In line with the validation rules for internal models, the authors consider historical quarterly data over a 5-year horizon (2009-2014) to build the internal models, and then validate these models on the observations from 2015.

The main advantage of the study by Fraisse and Laporte (2022) is that it exactly replicates the IRB methodology for a European bank. Indeed, in the first step, the authors construct a continuous risk score expressed as a function of ten risk factors using either logistic regression or an ML algorithm (Random Forest, Boosting, Ridge regression, and ANN). In the second step, they group firms into rating grades according to constraints in terms of (i) risk score homogeneity within a grade and (ii) risk score heterogeneity between grades. To do so, the authors employ an algorithm designed to define the best rating scale with respect to risk heterogeneity Z-tests. Finally, in the third step, the authors assign a PD to each grade of the final rating, defined as the 1-year average default rate, and then calculate the capital requirements according to the Basel formula associated with the IRB Foundation (IRB-F) approach.

The comparison between traditional and ML models is particularly relevant given the fact that the authors faithfully replicate the entire preprocessing chain implemented by banks before applying a risk-differentiation model. These preprocessing steps notably include eliminating variables with too many missing values, controlling for correlations between variables, discretizing continuous variables, and creating a class associated with missing values. The selection of model variables is based on both a quantitative approach (univariate logistic regressions) and a qualitative approach (interviews with Banque de France agents in charge of rating companies). The comparison of traditional and ML models is based on three criteria: (i) predictive gains measured in terms of AUC and F-score, (ii) the ability to pass the compliance tests used by supervisors during on-site missions for internal model validation, and (iii) the induced changes in required capital, measured by the ratio of RWA

[9] The data were drawn from the risk center (credit register) and the FIBEN database (FIchier Bancaire des ENtreprises). They cover corporate credits excluding companies belonging to the financial, real estate, public, and non-profit sectors. The database includes quarterly observations on $229,657$ companies from March 2009 to June 2015, totaling $5,846,627$ observations.

to the total assets of the bank.[10]

The results in Fraisse and Laporte (2022) are generally consistent with those in the academic literature regarding predictive performance. Indeed, Random Forests outperform the traditional parametric model in predicting corporate defaults, as well as other ML techniques (XGBoost, Ridge regression, and ANN). The latter delivers approximately the same AUC and F-scores as logistic regression once preprocessing is accounted for. Thus, the average AUC measures obtained on the six credit portfolios increase from 83% for logistic regression to 87% for Random Forests, while the differences with other ML techniques are smaller, which is in line with the findings of Thomas (2000), Baesens et al. (2003) et Lessmann et al. (2015).

In addition to predictive performance, the study reveals notable differences in the ability of ML models to pass validation tests and reduce capital requirements. Some ML algorithms lead to an increase in RWA, compared not only to the logistic model but also to the standardized approach. In contrast, Random Forests lead to significant reductions in RWA, but only for a subset of banks. Moreover, Random Forests and Gradient Boosting generally fail validation tests outside the model calibration period. On the validation and RWA dimensions, the most promising results are obtained for the ANN method, which pass compliance tests and lead in some cases to a significant reduction in capital requirements. Specifically, ANNs lead to a reduction in RWA ranging from 2% to 27% compared to logistic regression models.

In the same vein, the study by Alonso and Carbó (2020) focuses on a consumer credit portfolio of a Spanish bank. The authors first show that Random Forests and XGBoost perform better than the traditional approach based on logistic regression in terms of both risk differentiation (AUC) and calibration (Brier score). The authors then seek to evaluate the gains in terms of regulatory capital using, for the risk differentiation phase, an XGBoost model instead of a Lasso penalized logistic regression. To do so, they replicate exactly the IRB methodology by constructing risk classes from the default probabilities estimated by the ML model, and then calibrate the default probabilities within each class on the historical frequency of observed defaults on a training base. Their results show that the gains in

[10]The ability of the models to pass compliance tests is measured in part by a risk differentiation indicator corresponding to the number of times the z-test leads to rejection, for each quarter of the test period, of the null hypothesis of equality of average default rates between two adjacent risk classes. The higher this indicator is, the better the rating scale, as it allows for better risk discrimination.

regulatory capital allowed by XGBoost reach an average of 12%, or even 17% in scenarios based on different correlation values. Unlike Fraisse and Laporte (2022), the authors do not apply the entire risk class validation procedure, most notably the tests for risk homogeneity within each class. Moreover, since the two models were calibrated, the overall average PD of each did not differ significantly. Therefore, the reduction in regulatory capital achieved using the XGBoost method can come either from a difference in the allocation of credit within the risk classes, or from the difference in the number of risk classes obtained from the calibration procedure. This study shows the important role of both effects in reducing the required regulatory capital.

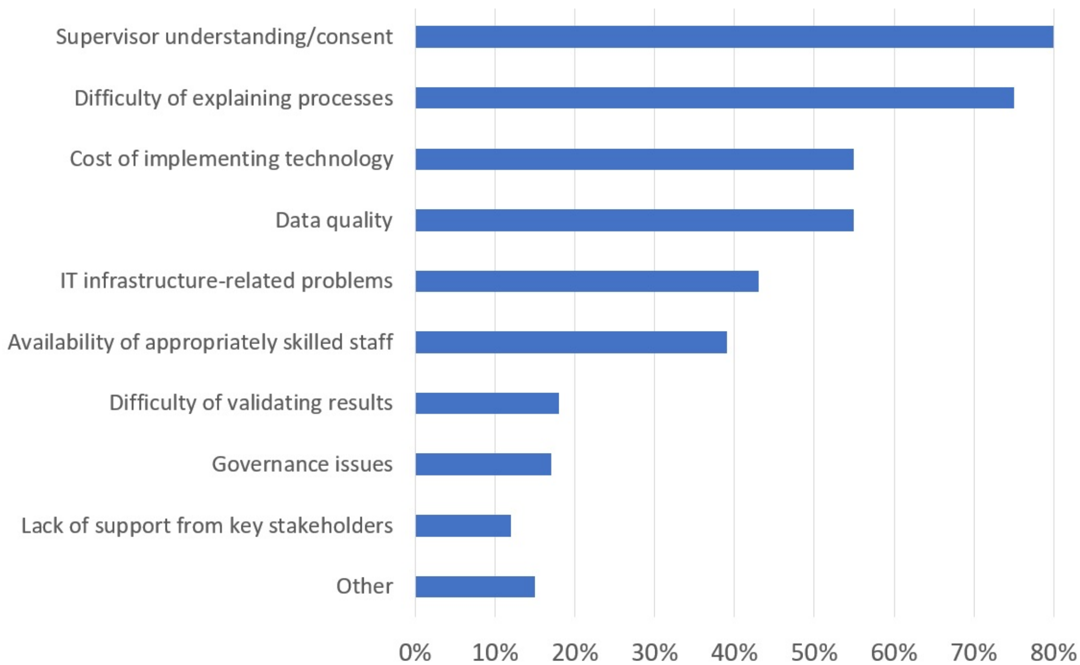
In a study conducted by FICO, Ould (2022) estimates a Gradient Boosting model for three loan portfolios (mortgage loans, consumer credit, and other loans) and compares the corresponding capital requirements to those reported by a European bank when using its IRB models. For this comparison, he considers input features identical to those on the initial scorecards. The results point towards a decrease in RWAs of 20.8% for consumer credits, 11.1% for loans, and 8.1% for mortgages. However, the study provides no information about model validation.

LGD modeling. For LGD, there is no analogous study on the potential economic gains from using ML techniques. However, Hurlin et al. (2018) evaluate various loss functions defined in terms of regulatory capital for LGD models. These loss functions, based on a quadratic error defined in terms of regulatory capital (calculated from the realizations of risk parameters), can be symmetric or asymmetric. They are designed to penalize any underestimation of the required regulatory capital. These loss functions enable the comparison of the predictive performance of different LGD models, including ML models, directly in terms of gains or losses in required capital. In an empirical application on a portfolio of auto loans, the authors compare various ML models (ANN, SVM, Random Forests) and parametric models (Fractional Response Regression model). The findings indicate that the ranking of models based on capital loss functions can differ substantially from those obtained based on standard statistical criteria (Mean Squared Error, Mean Absolute Error). This again illustrates the potential divergence between statistical accuracy gains and economic gains from ML.

5 The challenge of explainability/interpretability

When surveyed about the main challenges they face when implementing ML in credit risk modeling (Figure 3), US and international banks first mention model interpretability/explainability, both for internal control and external authorities. In the specific context of IRB models, the first EBA discussion paper raise the same concerns (EBA, 2021). The question of interpretability and its definition by the regulator will undoubtedly condition the future adoption of ML in the IRB context.

Figure 3: Main challenges for ML in the credit scoring



Source: IIF (2019)

5.1 Which explainability/interpretability for IRB models?

The issue of model interpretability is a central concern in all high-risk applications of ML and AI (EC, 2021). This problem arises because ML models can be highly complex, making it difficult, or even impossible, to interpret their predictions and understand the factors contributing to these predictions. Clearly, in the IRB context, it is not feasible to entrust the determination of a bank regulatory capital to a black box model. The supervisor, internal validation services, and the bank's management team must be able to ensure that the risk parameters used for calculating capital requirements accurately reflect the level of risk taken by the institution. As a result, IRB models must be interpretable and understandable by all stakeholders. However, the most effective ML models for credit risk management, such as Random Forests and gradient boosting methods, are non-interpretable and are typically considered black boxes, which consequently limits their use in the IRB context.

We need to distinguish here between the concepts of explainability and interpretability (EC, 2020). Interpretability refers to the degree to which a human being can understand how the model works and how it makes its decisions. Explainability, in contrast, refers to the possibility of providing a comprehensible explanation for the decisions the model makes. In other words, interpretability focuses on a global understanding of the model, while explainability focuses on a local understanding of individual decisions made by the model. In the IRB context, these two levels of analysis do not address the same stakeholders. Thus, according to Bracke et al. (2019), first-level modelers and validators must be interested in both interpretability and explainability to understand not only the essential factors that determine risk parameters, but also what determines the PD and LGD forecasts obtained for each contract. Conversely, supervisors and second-level validators should be more concerned with interpretability, i.e., the general operation of the model, and especially its ability to measure risks correctly in a new economic environment.

5.2 Post hoc interpretability/explainability methods

When models derived from ML algorithms are not interpretable a priori, the usual approach involves implementing methods to make them interpretable ex post (Molnar, 2019). These methods are said to be "model-agnostic" as they can be used to interpret the results of any ML model. The general idea of these methods is to perform reverse engineering based on inputs (predictors) and outputs (predictions) alone to reveal the algorithm's decision rule. Below, we distinguish between global methods intended to address interpretability concerns, and local methods designed to address explainability concerns.

Global methods. A simple global method consists of constructing a surrogate model that is natively interpretable (e.g., logistic regression, decision tree). The idea is to provide an approximation of the non-interpretable decision rule of an ML algorithm through a simpler and interpretable surrogate model. The latter is considered a global approximation of the true decision rule. However, the quality of the interpretation depends fundamentally on the goodness of fit of the surrogate model.

Other approaches based on graphical analyses do not need surrogate models. One such approach is that of the Partial Dependence Plots (PDP) (Friedman, 2001). Constructing PDPs involves scanning a set of possible values for an explanatory variable of interest, applying the ML predictive model using these successive values for all individuals (while keeping the values of other explanatory variables unchanged), and calculating, for each value tested, the average of the predicted outcomes. The PDP displays the marginal effect of a feature on the prediction. This plot reveals the nature of the relationship between the target and a predictor (e.g., flat, increasing, linear, or monotonic). However, the PDP relies on the strong assumption of independence between model features. Several recent extensions of PDP, such as the Accumulated Local Effects (ALE) plots (Apley and Zhu, 2020), have enabled the relaxing of this assumption of independence.

Local methods. Local methods provide an explanation for a decision made by the black box model at the individual level, such as that of a borrower. These methods aim to explain, for example, why the model has estimated the one-year PD of a given company at 4%. There are many local explanation methods but LIME (Ribeiro et al., 2016) and SHAP (Lundberg and Lee, 2017) are the most popular among banks.

The intuition of the LIME method (Local Interpretable Model-Agnostic Explanations) is to create local prediction models for each individual, which are simpler and easier to interpret than the original model. For a given reference individual, we create perturbed samples by randomly introducing small changes or noise in his features. Then, the initial black box model is applied to these simulated data to obtain new predictions. Finally, we estimate an interpretable model from the simulated characteristics and the new associated predictions (weighted according to their distance to the reference individual). This is typically a decision tree or a Lasso type penalized regression. This interpretable model permits us to approximate the influence of each feature on the final decision of the initial black box model, for the reference individual. Thus, for example, the bank can explain why a certain borrower has been classified as high-risk and which characteristics led to this choice.

The second local interpretation method widely used by banks is SHAP (SHapley Additive exPlanations). This method is based on the Shapley decomposition (Shapley, 1953), which allows for the fair distribution of a payout among players. In the context of interpretable ML, the model's prediction obtained for an individual is akin to the "payout" that we aim to distribute among the different model features, which correspond to the "players".

Shapley values are supposed to quantify the "additional profit generated by each player". In the context of ML, this profit is defined as the difference between the prediction for the reference individual and the average of predictions for the other individuals. Thus, the Shapley contribution of a feature corresponds to its marginal and weighted contribution to the prediction. This contribution is calculated from all possible combinations in which this feature has been added to the set of other features. This decomposition has the advantage of being based on a theoretical foundation and displays several interesting properties.

Calculating Shapley values becomes cumbersome as soon as the number of features in the model grows large. This is due to the large number of possible combinations of features to consider. To deal with this problem, Lundberg and Lee (2017) developed the SHAP method to approximate Shapley values in high-dimensional problems. However, despite these efficient calculation methods, computing Shapley values can still be costly in terms of computing time, depending on the number of features, observations, and complexity of the model under consideration.

Several extensions of these local methods have been developed, particularly for credit risk models. For instance, Bracke et al. (2019) propose a comprehensive approach based on Shapley values, called Quantitative Input Influence (QII), through which they derive a measure of the global influence of a variable on the model forecasts. This graphical representation, similar in spirit to PDP, allows for consideration of both non-linearities and interactions between variables.

Shortcomings of post hoc interpretability methods. The approach of using a black box ML model for risk differentiation, and then implementing a global or local interpretability/explainability method ex post is highly debated. For instance, Rudin (2019) calls for a complete cessation of the use of black box models for high-risk applications, even when these models are accompanied by interpretable ML methods. The first reason is that post hoc interpretability methods provide explanations that are not faithful to the original model calculations. The commonsense argument is that if the explanation were perfectly faithful to the original model, the explanation would be identical to the latter and, therefore, the original model itself would by definition be interpretable. Therefore, there is a risk that post hoc interpretability methods provide an inaccurate representation of the original model in certain parts of the feature space. The second reason is that explanations are sometimes nonsensical or provide insufficient detail to understand how the model works. Even when the original ML model produces a correct prediction and the interpretability method provides a good approximation of this prediction, it is possible that this explanation may omit so much information that it will ultimately make no sense. The third reason is that the explanations provided by different interpretability methods for the same model are sometimes inconsistent (Krishna et al., 2022). When multiple methods give different explanations, it is not clear which one to trust.

A recent article by Chen et al. (2022) also highlights the instability of interpretations provided by LIME and SHAP in the case of a low default portfolio. In ML, the problem of imbalanced data is well known, as it both influences the learning capacity of the models (the prediction of the dependent variable tends to be biased in favor of the majority class) and invalidates some evaluation criteria (e.g., the percentage of correct classifications). However, this configuration is often found in PD modeling, because observed default rates are usually very low. Several methods based on resampling techniques (SMOTE, ROSE, etc.) have been proposed to improve the classification performance of ML models in the presence of imbalanced data. However, Chen et al. (2022) show that data imbalance also affects interpretability methods.

5.3 Is there truly an interpretability/performance trade-off?

According to a commonly accepted idea, complex black box ML models provide more accurate risk-parameter estimates. Because of this overperformance, it is rational for banks to accept the use of such opaque models, and then to apply post hoc interpretability methods, regardless of their shortcomings. However, this alleged trade-off between performance and interpretability is increasingly being challenged in the academic literature. For instance, Semenova et al. (2022) base their argument on the concept of the Rashomon set, which is defined as the set of predictive models that deliver similar accuracy to the optimal black box model. Since the data are finite, there are many different models, with different functional forms and predictions, which perform almost as well as the best model. In other words, a large Rashomon set exists.[11] Thus, in most use cases, the Rashomon set is probably sufficiently large to include interpretable models. If this theory is verified, models exist that are both accurate and interpretable.

[11] In practice, it is often observed that many different supervised ML classification algorithms provide relatively close results in terms predictive performance on the same dataset, even though they have very different specifications.

If we admit that, *theoretically*, there may not be a trade-off between performance and interpretability, we still need to find *in practice* some banking scoring models that are both efficient and natively interpretable. Fortunately, several recent studies have shown that this is achievable in practice. For instance, Dumitrescu et al. (2022) propose a hybrid approach, called PLTR, based on a logistic regression model including predictors extracted from decision trees. These predictors correspond to binary rules (leaves) produced by very shallow decision trees built with the original predictive variables. To handle a potentially high number of decision tree rules, the logistic regression model is estimated through an adaptive Lasso-type penalized regression that automatically selects the most important predictors. This variable preprocessing phase thus improves the predictive performances of a simple logistic regression by capturing potential non-linear effects, such as threshold effects and interactions between explanatory variables, while maintaining its interpretability. Its application to a credit portfolio shows that the PLTR model can predict defaults as well as a Random Forest.

Along similar lines, Flachaire et al. (2022) propose the GAM(L)A approach based on generalized additive models (GAM). This model relies on parametric and non-parametric functions of the initial variables, allowing to capture linearity and non-linearity between dependent and explanatory variables. It also includes a variable selection procedure to control for overfitting (Lasso or Autometrics type). Their empirical analysis based on a consumer credit portfolio show similar predictive performance for this natively interpretable model to those of Random Forest or XGBoost models. We can also mention recent models such as Explainable Boosting Machine (EBM), which are similar to GAM models, but automatically detect and include pairwise interaction terms (Nori et al., 2019), or GAMI-Net, which are GAM models with structured interactions (Yang et al., 2021). Most of these methods are available on PiML (Python Interpretable Machine Learning), a comprehensive Python toolbox designed to facilitate the development and evaluation of interpretable machine learning models.

To illustrate how deeply the myth of the trade-off between accuracy and interpretability is ingrained in the minds of credit scoring professionals, Semenova et al. (2022) highlight the example of the interpretable ML challenge, which was organized in 2018 by FICO. In this challenge, the task was to create a black box ML model to predict defaults on a credit portfolio and then explain the model's forecasts using interpretable ML techniques.

Ultimately, a globally interpretable two-level GAM (Chen et al., 2018) won the competition. The design of the FICO challenge clearly illustrates that the organizers did not expect an interpretable model to win, as they had not even asked the participants to try to build such a model. However, this new approach of natively interpretable ML is likely to develop within banks in the coming years, as indicated for example by the research currently being conducted by Wells Fargo on the matter (Sudjianto and Zhang, 2021).

6 Other important challenges

Additional challenges can arise when applying ML within the IRB framework. These challenges can be categorized into two main groups: (i) issues related to ML governance, and (ii) concerns associated with operational risks.

6.1 Governance

The growing adoption of AI and ML by banks has prompted regulators to suggest best practice guidelines for the management and governance of these innovative technologies (ACPR, 2020; EC, 2020). For instance, the EBA (2020) identifies four essential pillars for the development, implementation, and adoption of AI in the banking sector. The first pillar focuses on data management and aims to ensure the use of quality data, while accounting for the constraints of personal data protection, particularly compliance with the General Data Protection Regulation (GDPR). The second pillar concerns the security and reliability of technological infrastructures (databases, computing infrastructures, software, etc.). The third pillar centers on training. As most of the ML methods currently used in banking were developed between the mid-1980s and early 2000s, graduate programs in economics and finance have only started to incorporate these methods into their curricula in recent years.[12] Therefore, the use of ML must be accompanied by a significant training effort aimed at those responsible for model construction and validation as well as at executives, board members, and banking supervisors. The final pillar is devoted to the methodology that should be established to facilitate the development and adoption of AI solutions within banks. The development of an ML project follows a lifecycle with specific stages, such as data preparation, modeling, validation, and monitoring over time. Governance should also aim

[12] For example, the CART algorithm for classification trees was published in 1984 (Breiman et al., 1984), bagging methods in 1996 (Breiman, 1996), and Random Forests in 2001 (Breiman, 2001). The only exception among popular methods in finance is the XGBoost method of Chen and Guestrin (2016), but the Boosting original idea was first published in 1997 by Freund and Schapire (1997).

to implement trustworthy AI, notably integrating the dimensions of ethics, explainability, interpretability, validation, traceability, and auditability (EBA, 2021).

6.2 Operational risks

An important source of operational risk in the implementation of ML is the risk of model overfitting, which is defined as a situation in which the model fits the training data excessively well (i.e., the bias of the forecasts is low) but cannot correctly generalize the forecasts to new data (i.e., the variance of the forecasts is high). Overfitting results in a greater prediction error on the test sample than on the training sample. In the IRB approach, controlling the risk of overfitting is essential because the *raison d'être* of internal models, which are trained on a training set of past defaults, is precisely to provide reliable forecasts of risk parameters for loans that have not yet defaulted (test set). Thus, overfitting can lead to an underestimation of the capital requirements.

Overfitting risk can be controlled by carefully choosing the value of the hyperparameters. The latter are configuration parameters that determine the general shape of the ML model. Hyperparameters should not be confused with the model parameters, which are instead determined by the algorithm to allow the model to fit the data well. In contrast, hyperparameters are not determined by the learning algorithm itself but are defined before the model is trained. For example, for a decision tree, the hyperparameters are the depth of the tree and the minimum number of observations per leaf. Once these hyperparameters are set, the algorithm determines the selected features and estimates the model parameters, i.e., the thresholds that define the decision rules at each node. Any change in the value of the hyperparameters can radically change the functional form of an ML model, its complexity, its level of interpretability, its decision rules, and ultimately its predictions. To make this risk more concrete, EBA (2021) uses the example of a classification tree. If the depth of the tree is set to two, a very simple model is obtained that can predict at most four different values, using information from at most three different features. However, if the depth is set to ten, the complexity of the model grows significantly as the number of possible different predictions climbs to 1,024 and the model can mobilize up to 1,023 different features. This example illustrates how small changes in the hyperparameter values, or in the way they are determined, can lead to significant changes in the structure of an ML model and its predictive capabilities.

Different methods allow us to set an optimal value for hyperparameters to control the trade-off between bias and variance and to limit overfitting risk. The best-known method is cross-validation and its variants, including hold-out, K-fold or leave-one-out. In general, these methods involve separating the data into a training sample and a validation sample. The optimal value of the hyperparameters is then sought by optimization (i.e., optimization on a grid of values, random optimization, or Bayesian optimization), to achieve the best model performance on the validation sample. Once the optimal hyperparameters have been determined to limit overfitting, the model is trained on all the observations in the training and validation databases. The final phase of model evaluation is carried out using a test sample, which is different from the training and validation samples.

Setting the hyperparameters always relies on a set of arbitrary choices left to the judgment of the modeler. These choices are a source of operational risk and must be strictly validated in the IRB context. The first choice is simply to decide which hyperparameters will be fine-tuned by cross-validation and which ones will be left at their default values in the statistical package (Python or R). Even though experienced developers often have a clear intuition about the appropriate hyperparameters for a given problem, it is important to ensure the validity of these choices. The second arbitrary choice concerns the value grids or limit values used for the search optimization. Finally, the size and quality of data can significantly affect the hyperparameter determination: The larger and more diverse the dataset, the more likely it is that the obtained hyperparameters will be robust and generalizable. Alternatively, when the dataset is limited in size or fails to reflect the diversity of the data, the selection of hyperparameters can result in poorly defined models. Therefore, due to the critical importance of setting the hyperparameters and the predominance of human judgment in this step, the EBA has underscored the need to verify these choices meticulously. Specifically, the experts responsible for validating internal models must check the logic underlying the choice of hyperparameters. The problem is that such verification can be particularly difficult, especially for complex models, as it requires a deep understanding of the methodology. This type of validation is also essential to limit regulatory arbitrage, which would consist, for instance, of adjusting the hyperparameters according to cross-validation approaches to reduce the level of risk parameters and, ultimately, capital requirements.

Recently, the academic literature has provided empirical evidence on the importance of operational risk in the context of credit risk management. For example, Fraisse and Laporte

(2022) show that homogeneous risk classes, established from the scores of certain ML models, fail validation tests due to the overfitting of these models. This is particularly the case for Random Forests, which display the highest level of predictive performance on the training set. Even though the hyperparameters of these models have been set according to a K-fold cross-validation approach aimed at limiting overfitting risk, the authors observe a nearly 10-point drop in the average AUC obtained on the six training bases and the six test datasets, and a 44-point drop in F-score. This instability explains the lack of robustness of the risk scores produced by the model.

Setting hyperparameters can also generate other types of operational risks. In the context of a bank scoring model, Hurlin et al. (2022) show that small modifications to the hyperparameters can have significant consequences in terms of fairness and the emergence of discrimination biases. More precisely, their study illustrates that a change in the grid of values used to optimize the hyperparameters of a classification tree leads to obtaining two score models with overall comparable predictive performances (AUC), but one of which generates significant gender discrimination in loan granting. Thus, even if gender is not included in the feature space of the model, a classification tree, depending on the value of the hyperparameters, can lead to gender discrimination, with gender being approximated by other legitimate variables in the decision rules.

7 Conclusion

Several conclusions and recommendations emerge from our survey. First, ML offers obvious advantages in the IRB context. Used upstream or downstream of the risk-differentiation phase, these methods deliver significant gains in productivity and efficiency for many important tasks, such as data preparation, variable transformation, and the construction of challenger models. During the central phase of risk differentiation, certain ML models, notably homogeneous ensemble models, generate more precise estimates of risk parameters (PD and LGD) than those provided by standard parametric approaches. However, these gains tend to plateau with increasing model complexity. Thus, according to the academic literature, it becomes increasingly difficult to significantly improve the performance of banking scoring models by developing new classification methods. Second, certain ML models utilized for risk differentiation permit decreases in regulatory capital by between 2% and 20%. Others comply with the validation criteria imposed by the banking supervisor. However, no ML method systematically stands out in both dimensions. In practice, the choice of ML algorithm jointly influences the probability of regulatory approval and the level of capital requirements.

The adoption of ML also raises several important challenges. The first is the acceptability of these methods, both by the supervisor and by the banks themselves. The acceptability level is high when ML is used upstream of risk differentiation or to validate internal models. However, the situation is quite different in the risk differentiation phase. How can a bank justify the use of a Random Forest or an XGBoost model, instead of the standard logistic regression model? On the one hand, if this change results in a slight reduction, or even an increase, in regulatory capital, the bank will have no interest in adopting ML and redesigning its internal risk models. On the other hand, if this change leads to a strong reduction in regulatory capital requirements, it could be associated with regulatory arbitrage by the supervisor. Regulators' suspicion will be particularly strong for models relying on hyperparameters that somewhat opaquely affect the model forecasts. In this case, only objective criteria for measuring accuracy gains in the estimates of risk parameters can solve this dilemma. However, even then, the academic literature indicates that ML models that lead to a better estimation of risk parameters in terms of typical statistical criteria (AUC, sensitivity, precision, etc.) are not always necessarily those that allow banks to reduce their RWA or that obtain supervisor approval.

The second challenge is that of the interpretability and explainability of ML models. It is evident that the calculation of a bank's regulatory capital cannot be delegated to a completely opaque model. In this context, the usual approach is to use non-interpretable ML models coupled with techniques to make their predictions and functioning interpretable ex post. The problem with such approach is that no universal explanation exists for the operation of a complex model. Moreover, there is a plethora of interpretability methods providing just as many explanations. The question then arises of what type of interpretations and explanations the supervisor expects for the workings of internal models. It is undoubtedly this uncertainty about the regulatory validation procedure of ML models that explains why these tools are still very rarely used by banks in the IRB or IFRS 9 contexts.

We show that one way out of this situation is to promote the use of natively interpretable ML models in the IRB context. Recent theoretical work and empirical studies in credit scoring have challenged the idea of a necessary trade-off between performance and interpretability. When used in the context of risk differentiation, interpretable ML models can provide risk parameter estimates that are as accurate as black box type models. Even if natively interpretable ML helps overcome the issues related to the lack of interpretability, however, the challenges of governance and operational risks remain and need to be taken seriously.

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